NODE.JS AUTHENTICATION WEB APPLICATION

Project Title: **Node.js Web Application with Authentication**
Course: **Web Programming (Seminar Homework)**

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Live URL: http://143.47.98.96/app207/login

login details:

login = admin@gmail.com

Password = admin123

Linux Server: ssh student207@143.47.98.96

Linux password: Stud207@2025

PROJECT DESCRIPTION

This seminar project focuses on creating a Node.js web application with authentication features such as Register, Login, and Dashboard. The application was built using Express.js, EJS templates, MySQL database (hosted on Railway), bcrypt for password hashing, and deployment on Render cloud hosting

The aim of this project is to demonstrate practical knowledge of:
✔ Server-side programming with Node.js
✔ Secure authentication using bcrypt

✔ **Database integration using MySQL**
✔ **Cloud deployment using Render**
✔ **Responsive UI using Tailwind CSS**

All steps of development are documented with explanations and screenshots across the following sections

This project is a practical implementation of a Node.js web application with user authentication features (Register, Login, Logout, and Dashboard). The main purpose of this work is to understand how server-side web development works using Express.js, EJS templating, MySQL cloud database (Railway), and deployment via Render.com.

In this documentation, I explain all the steps from installation of Node.js, creating a project folder, installing necessary packages, connecting to a cloud database, building authentication (with bcrypt for password hashing), and deploying the final application online. All screenshots are included to show the real implementation on a live server.

The aim of this task is to demonstrate:

- How to create a **Node.js** server using **Express.js**

- How to design dynamic frontend pages using **EJS**

- How to connect Node.js with **MySQL** database on **Railway**

- How to implement secure authentication using **bcrypt**

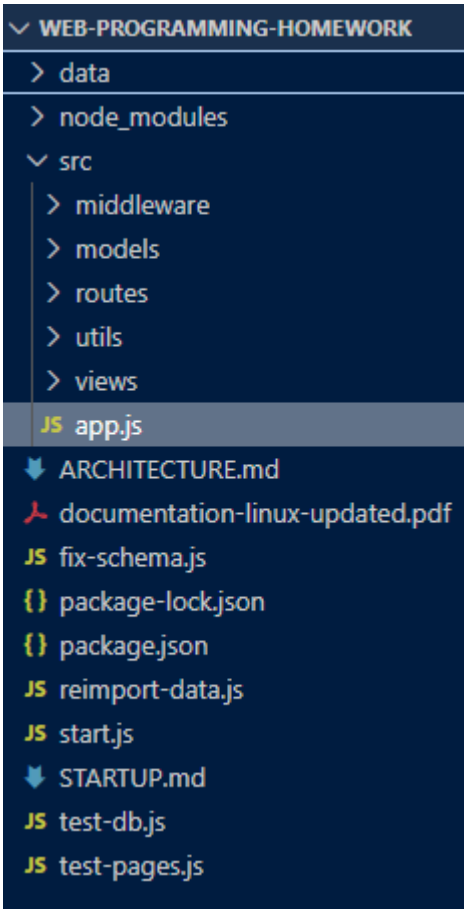
- How to deploy and host a live web application using Render.com

This project is useful for learning full-stack web development, cloud deployment, and backend programming with JavaScript. It also shows how to solve real-world problems such as database connection errors, deployment challenges, authentication logic, and responsive UI design.

Deployment to render screenshot

```
$ node exercise/start.js
Checking if tables are empty...
Users: 2, Students: 429, Subjects: 20, Marks: 17182
Tables are already populated. Skipping import.
Server running on internal port 4207
```

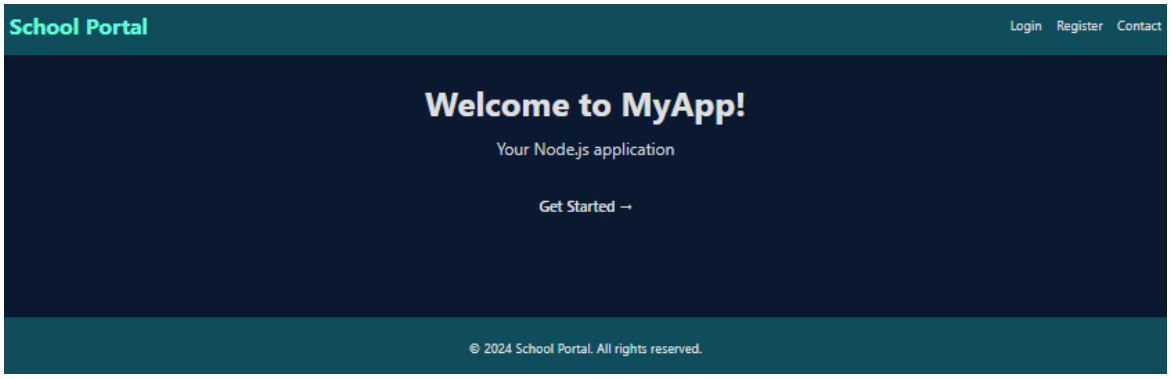
File structure



Home Page

Link: http://143.47.98.96/app207/

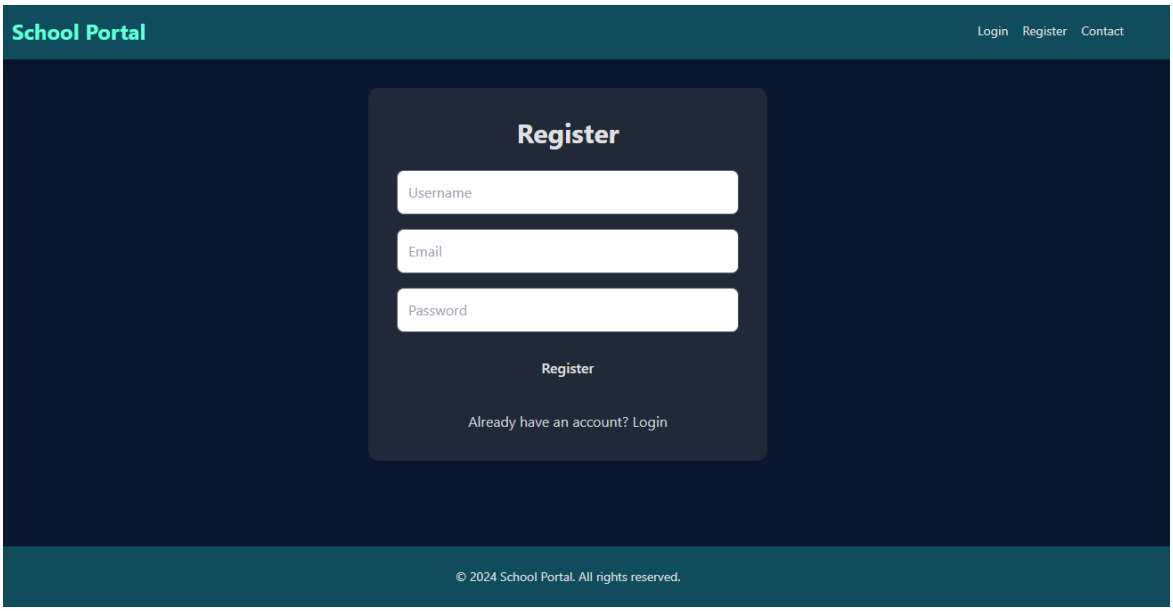
The Home screen serves as the "spectacular" front-facing introduction to the application, fulfilling the Mainpage menu requirement. It acts
as the primary landing page, welcoming visitors with a bold, responsive layout that establishes the brand identity immediately upon arrival. Designed to guide user flow, it features a prominent call-to-action that directs new users toward the "Get Started" registration process, effectively converting casual visitors into registered members of the platform.



Register Page

Link: http://143.47.98.96/app207/login

The Register page is the entry point for new users, fulfilling the mandatory registration requirement of the authentication system. It features a secure form that collects a username, email address, and password. On submission, the system cryptographically hashes the password using bcrypt before storing the new user record in the users table. This process automatically assigns a default role to the new account, enabling immediate participation in the platform's community features while maintaining the security integrity of the role-based access control system.



Database table (for storing user details)

Students Table (students)

This table serves as the primary registry for all student data. It is designed with a simple yet effective schema containing a unique student_id as the primary key and a name column. This structure allows for efficient identification and retrieval of student records, acting as a fundamental entity for linking academic performance data across the system.

←T→				id	student_id	name	created_at
☐	Edit	Copy	Delete	1288	1	Pék Roland	2025-12-15 21:10:21
☐	Edit	Copy	Delete	1289	2	Illin Zita	2025-12-15 21:10:21
☐	Edit	Copy	Delete	1290	3	Ördögh Dániel	2025-12-15 21:10:21
☐	Edit	Copy	Delete	1291	4	Barta Ildikó	2025-12-15 21:10:21
☐	Edit	Copy	Delete	1292	5	Simony Kata	2025-12-15 21:10:21
☐	Edit	Copy	Delete	1293	6	Csepreg Natália	2025-12-15 21:10:21
☐	Edit	Copy	Delete	1294	7	Nagy Petra	2025-12-15 21:10:21
☐	Edit	Copy	Delete	1295	8	Vörös Dóra	2025-12-15 21:10:21
☐	Edit	Copy	Delete	1296	9	Varga Boldizsár	2025-12-15 21:10:21
☐	Edit	Copy	Delete	1297	10	Szilágyi Réka	2025-12-15 21:10:21
☐	Edit	Copy	Delete	1298	11	Helle Ágnes	2025-12-15 21:10:21
☐	Edit	Copy	Delete	1299	12	Gyömbér Anna	2025-12-15 21:10:21
☐	Edit	Copy	Delete	1300	13	Farkas Dária	2025-12-15 21:10:21
☐	Edit	Copy	Delete	1301	14	Barta Zsuzsanna	2025-12-15 21:10:21
☐	Edit	Copy	Delete	1302	15	Schiss Bence	2025-12-15 21:10:21
☐	Edit	Copy	Delete	1303	16	Szabó Attila	2025-12-15 21:10:21
☐	Edit	Copy	Delete	1304	17	Nagy Dorottya	2025-12-15 21:10:21
☐	Edit	Copy	Delete	1305	18	Lengyel Lilla	2025-12-15 21:10:21
☐	Edit	Copy	Delete	1306	19	Csiszár Nikolett	2025-12-15 21:10:21
☐	Edit	Copy	Delete	1307	20	Kond Lili	2025-12-15 21:10:21
☐	Edit	Copy	Delete	1308	21	Fehér Orsolya	2025-12-15 21:10:21
☐	Edit	Copy	Delete	1309	22	Haskó Fanni	2025-12-15 21:10:21
☐	Edit	Copy	Delete	1310	23	Sós Gábor	2025-12-15 21:10:21
☐	Edit	Copy	Delete	1311	24	Kondász Ferenc	2025-12-15 21:10:21
☐	Edit	Copy	Delete	1312	25	Magyari Tekla	2025-12-15 21:10:21

Subjects Table (subjects)

The Subjects table catalogs the academic courses available in the curriculum. It stores essential details including a unique id, the descriptive subject_name, and a subject_code (or category) for classification. This table acts as a reference point for the application, ensuring that every mark recorded is associated with a valid and defined course.

←T→				id	subject_name	subject_code	created_at
☐	Edit	Copy	Delete	1	Information Technology	information technology	2025-12-15 21:10:21
☐	Edit	Copy	Delete	2	Earth and Environmental Science	science	2025-12-15 21:10:21
☐	Edit	Copy	Delete	3	Literature	humanities	2025-12-15 21:10:21
☐	Edit	Copy	Delete	4	History	humanities	2025-12-15 21:10:21
☐	Edit	Copy	Delete	5	Mathematics	mathematics	2025-12-15 21:10:21
☐	Edit	Copy	Delete	6	Physics	science	2025-12-15 21:10:21
☐	Edit	Copy	Delete	7	Singing and Music	arts	2025-12-15 21:10:21
☐	Edit	Copy	Delete	8	Drawing and Visual Culture	arts	2025-12-15 21:10:21
☐	Edit	Copy	Delete	9	Philosophy	humanities	2025-12-15 21:10:21
☐	Edit	Copy	Delete	10	Biology	science	2025-12-15 21:10:21
☐	Edit	Copy	Delete	11	Chemistry	science	2025-12-15 21:10:21
☐	Edit	Copy	Delete	12	English Language	languages	2025-12-15 21:10:21
☐	Edit	Copy	Delete	13	French Language	languages	2025-12-15 21:10:21
☐	Edit	Copy	Delete	14	German Language	languages	2025-12-15 21:10:21
☐	Edit	Copy	Delete	15	Italian Language	languages	2025-12-15 21:10:21
☐	Edit	Copy	Delete	16	Russian Language	languages	2025-12-15 21:10:21
☐	Edit	Copy	Delete	17	Spanish Language	languages	2025-12-15 21:10:21
☐	Edit	Copy	Delete	18	Latin Language	languages	2025-12-15 21:10:21
☐	Edit	Copy	Delete	19	Hungarian Language	humanities	2025-12-15 21:10:21
☐	Edit	Copy	Delete	20	Arts	arts	2025-12-15 21:10:21

Marks Table (marks)

As the core relational table of the application, the Marks table links students to subjects and records their academic performance. It utilizes foreign keys (student_id and subject_id) to establish relationships with the Students and Subjects tables, respectively, while storing the actual score in the mark column. This design enables complex queries, such as generating student report cards or analyzing subject performance trends.

←T→				id	student_id	subject_id	mark	created_at
☐	Edit	Copy	Delete	51550	140	13	4	2025-12-15 21:10:21
☐	Edit	Copy	Delete	51551	105	13	5	2025-12-15 21:10:21
☐	Edit	Copy	Delete	51552	105	13	4	2025-12-15 21:10:21
☐	Edit	Copy	Delete	51553	345	6	5	2025-12-15 21:10:21
☐	Edit	Copy	Delete	51554	390	6	5	2025-12-15 21:10:21
☐	Edit	Copy	Delete	51555	365	6	5	2025-12-15 21:10:21
☐	Edit	Copy	Delete	51556	351	5	5	2025-12-15 21:10:21
☐	Edit	Copy	Delete	51557	366	3	5	2025-12-15 21:10:21
☐	Edit	Copy	Delete	51558	365	3	5	2025-12-15 21:10:21
☐	Edit	Copy	Delete	51559	360	3	5	2025-12-15 21:10:21
☐	Edit	Copy	Delete	51560	83	18	5	2025-12-15 21:10:21
☐	Edit	Copy	Delete	51561	419	19	5	2025-12-15 21:10:21
☐	Edit	Copy	Delete	51562	408	19	5	2025-12-15 21:10:21
☐	Edit	Copy	Delete	51563	140	13	5	2025-12-15 21:10:21
☐	Edit	Copy	Delete	51564	183	13	5	2025-12-15 21:10:21
☐	Edit	Copy	Delete	51565	155	13	5	2025-12-15 21:10:21
☐	Edit	Copy	Delete	51566	113	13	4	2025-12-15 21:10:21
☐	Edit	Copy	Delete	51567	129	13	5	2025-12-15 21:10:21
☐	Edit	Copy	Delete	51568	103	13	5	2025-12-15 21:10:21
☐	Edit	Copy	Delete	51569	150	13	5	2025-12-15 21:10:21
☐	Edit	Copy	Delete	51570	172	13	5	2025-12-15 21:10:21
☐	Edit	Copy	Delete	51571	297	13	4	2025-12-15 21:10:21
☐	Edit	Copy	Delete	51572	297	13	5	2025-12-15 21:10:21
☐	Edit	Copy	Delete	51573	262	13	4	2025-12-15 21:10:21
☐	Edit	Copy	Delete	51574	253	13	5	2025-12-15 21:10:21

Users Table (users)

This table manages authentication and authorization for the system. It securely stores user credentials, including username, email, and hashed password, along with a role column (e.g., 'visitor', 'registered', 'admin'). This role-based structure is critical for controlling access to protected features like the Dashboard, Messages, and Admin menus.

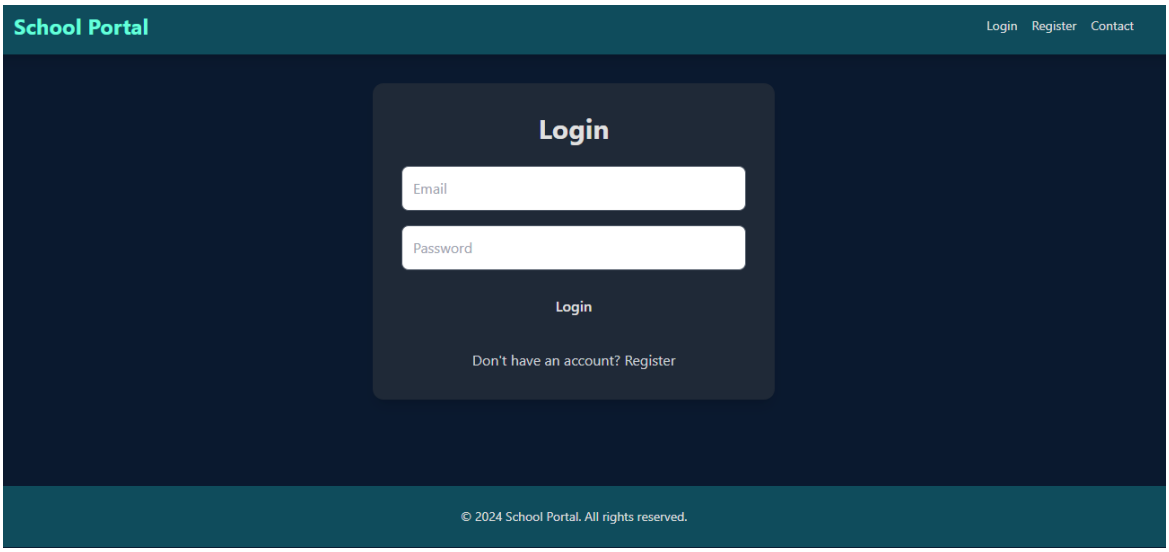
id	username	email	password	role	created_at
1	ayman	ayman@gmail.com	$2b$10$TPCkNXXQLoUTBr9RcG.RAu.BFgRmR77ePecvYqzDUx3...	user	2025-12-15 20:54:50
2	Admin	admin@gmail.com	$2b$10$Hm/BJastwbRuO8jz9nrlW.LgpNvf14dKbPEokxKhaKK...	admin	2025-12-15 20:54:50

Messages Table (messages)
The Messages table captures communication initiated through the Contact page. It records the sender's name, email, and the content of their message, along with a created_at timestamp. This table ensures that all user inquiries

id	name	email	message	created_at
1	ayman	ayman@gmail.com	Hello	2025-12-15 20:57:47

Login page

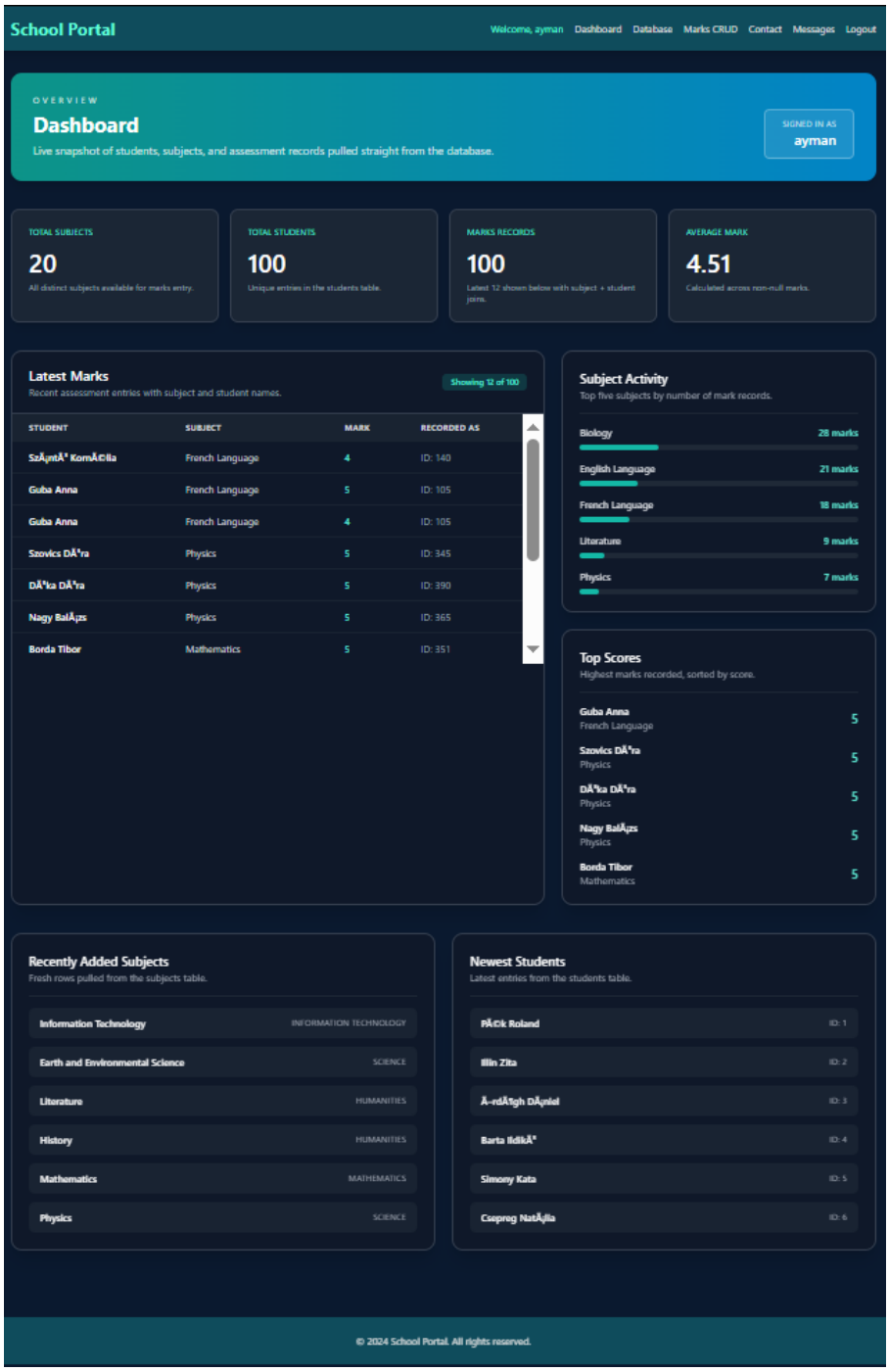
Link: http://143.47.98.96/app207/login
The Login page is a critical component of the mandatory authentication system, serving as the gateway to the application's protected features. It presents a secure form where users enter their email and password to authenticate their identity. Upon successful verification against the users table, the system establishes a secure session and redirects the user based on their role—granting "Registered Visitors" access to the Messages menu and "Admins" exclusive entry to the Admin dashboard—thereby enforcing the project's strict role-based access control requirements.



Dashboard screen

Link: /dashboard
The Dashboard serves as the central analytics hub, aggregating real-time data from the Students, Subjects, and Marks tables to provide a comprehensive system overview. It features dynamic visualizations of key metrics, including total record counts,

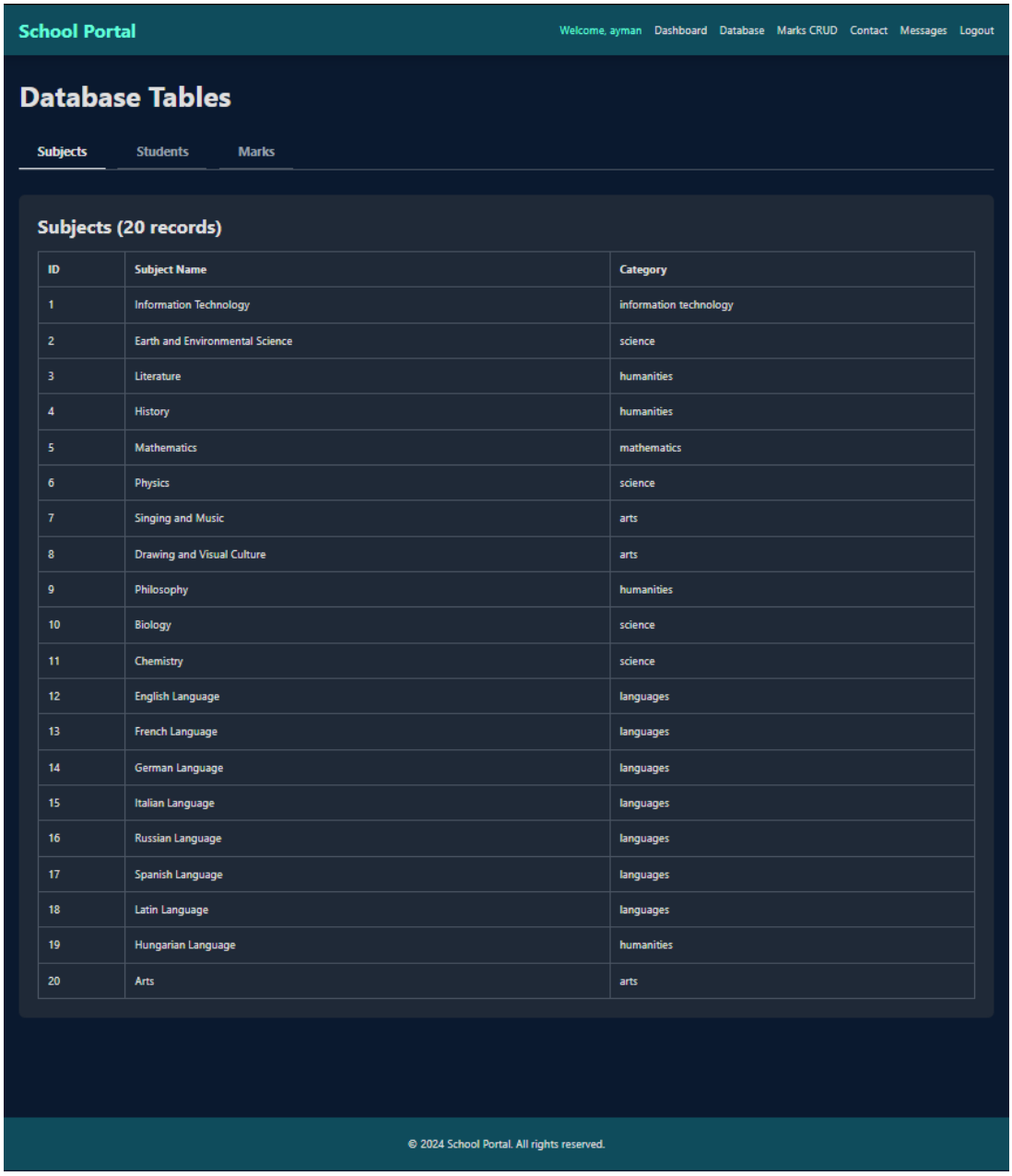
top-performing subjects, and a leaderboard of recent high scores. This page fulfills the requirement to display complex, multi-table data by using SQL joins and aggregation to present raw database entries as meaningful, human-readable insights for authenticated users.



Database screen

Link: http://143.47.98.96/app207/database
This page fulfills the requirement to display data from three distinct database tables: Students, Subjects, and Marks. It features a dynamic tabbed interface that allows users to seamlessly switch between different data sets without reloading the page. The

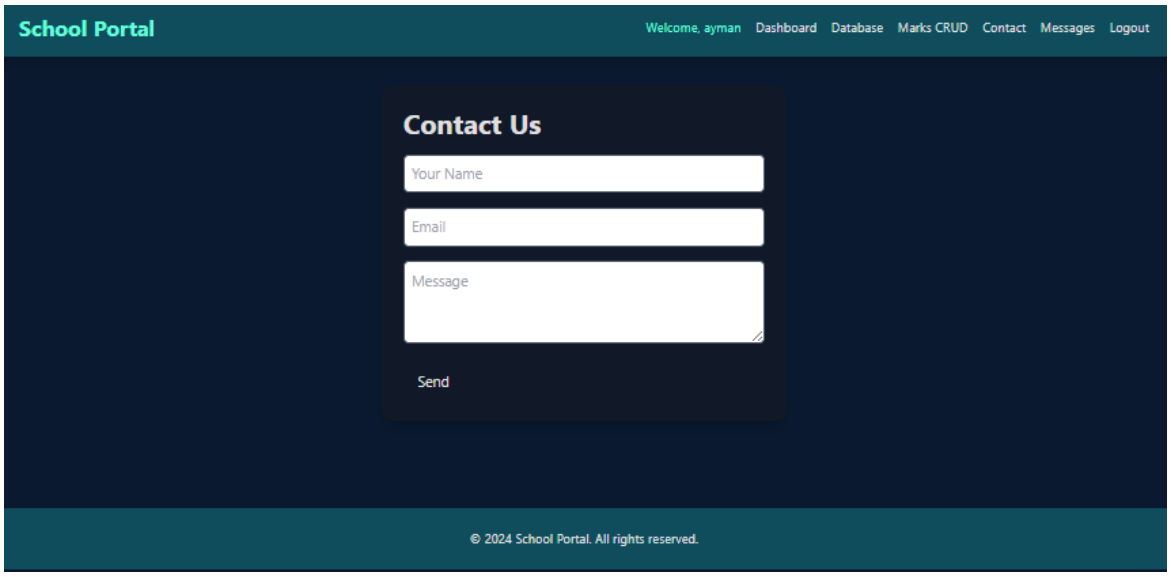
Marks view implements SQL joins to present human-readable student and subject names instead of raw IDs, providing a clear and comprehensive view of the underlying academic records. Access to this sensitive data is restricted to authenticated users only.



ID	Subject Name	Category
1	Information Technology	information technology
2	Earth and Environmental Science	science
3	Literature	humanities
4	History	humanities
5	Mathematics	mathematics
6	Physics	science
7	Singing and Music	arts
8	Drawing and Visual Culture	arts
9	Philosophy	humanities
10	Biology	science
11	Chemistry	science
12	English Language	languages
13	French Language	languages
14	German Language	languages
15	Italian Language	languages
16	Russian Language	languages
17	Spanish Language	languages
18	Latin Language	languages
19	Hungarian Language	humanities
20	Arts	arts

Contact menu

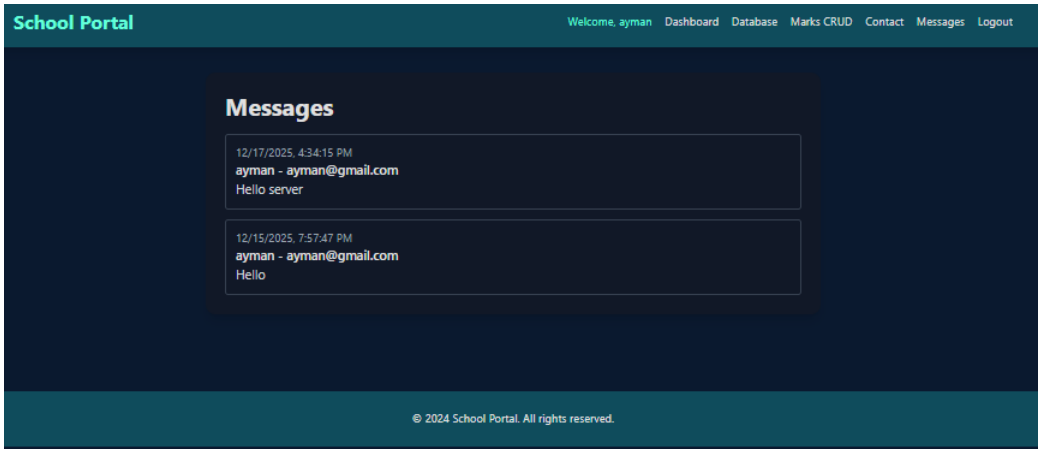
Link: http://143.47.98.96/app207/contact
The Contact page provides a direct communication channel for users to reach the site owner. It features a responsive form that collects the user's name, email, and message. Upon submission, the data is securely persisted to the database in the messages table, fulfilling the requirement to save form data. This feature is accessible to all visitors, ensuring open communication, while the backend includes a robust fallback mechanism to file storage if the database is temporarily unavailable.



Messages menu

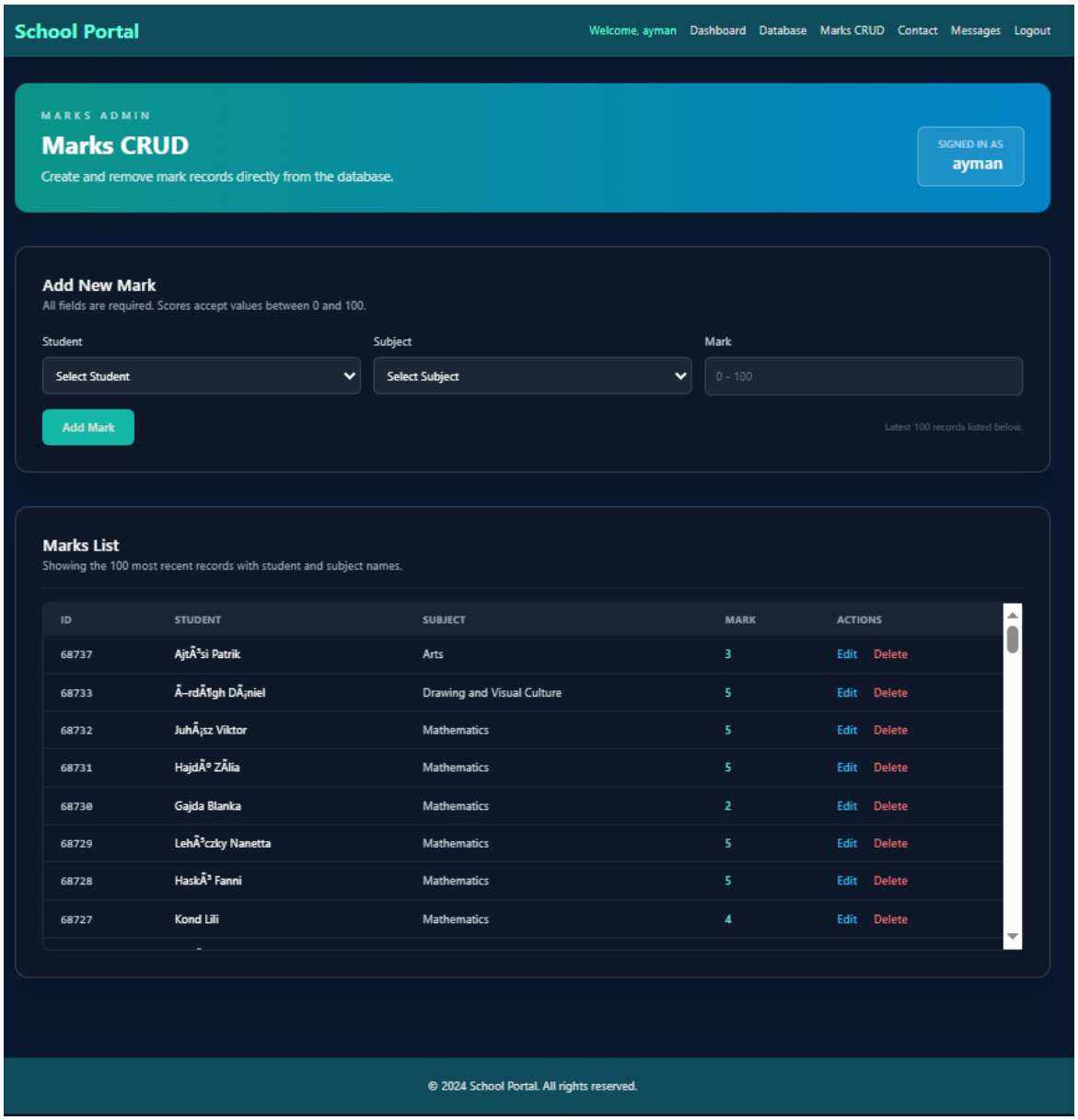
Link: http://143.47.98.96/app207/messages
This page displays a comprehensive list of all inquiries submitted through the Contact form, fulfilling the requirement to present database entries in descending chronological order (newest first). Each entry details the sender's name, email,

message content, and the precise timestamp of submission. Access to this sensitive information is strictly protected by authentication middleware, ensuring that only logged-in users (such as Registered Visitors and Admins) can view the communication history



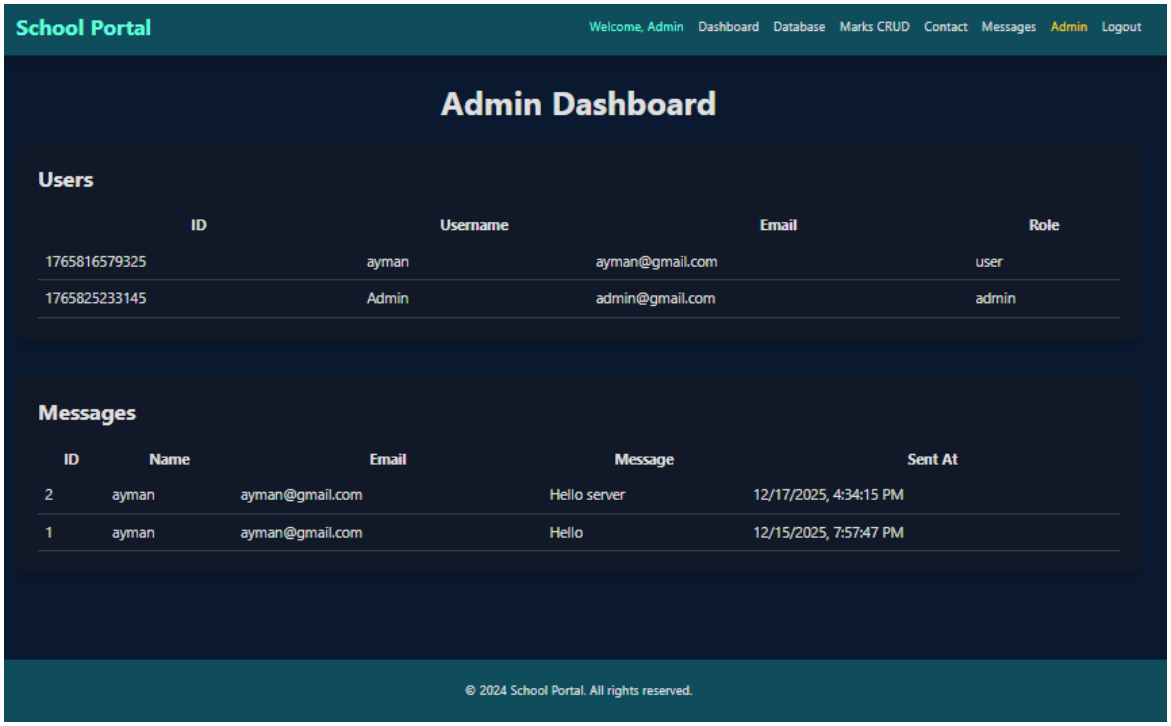
CRUD menu

Link: http://143.47.98.96/app207/crud
This page implements a full Create, Read, Update, and Delete (CRUD) application for the marks table, fulfilling the core requirement for database manipulation. It provides a user-friendly interface where authenticated users can add new academic records by selecting students and subjects from dropdown lists, view existing marks in a detailed table, and easily edit or delete specific entries. The backend handles these operations with secure SQL queries, ensuring data integrity while providing immediate visual feedback on the frontend.



Admin menu

Link: http://143.47.98.96/app207/admin
The Admin page is a restricted area accessible exclusively to users with the "admin" role, fulfilling the mandatory requirement for role-based access control. It provides a comprehensive overview of the system's user base and communication logs, displaying a list of all registered users alongside their roles and a complete history of messages received through the contact form. This centralized dashboard empowers administrators to monitor site activity and manage user data effectively, ensuring secure and efficient system administration.



Conclusion

This project successfully demonstrates the implementation of a robust, full-stack web application that meets all specified requirements, including secure authentication, role-based access control, and complex database interactions. By leveraging Node.js, Express, and MySQL, the system delivers a responsive and dynamic user experience, from the public-facing Home and Contact pages to the restricted Admin and CRUD interfaces. The seamless integration of multiple database tables (students, subjects